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Paleomagnetic Results on Early Tertiary Lava Flows from West Greenland and their bearing on the Evolution History of the Baffin Bay–Labrador Sea Region

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Paleomagnetic investigations have been carried out on about 250 oriented block samples collected from Early Tertiary lava flows representing an altitude span of about 1200 m on the Disko Island and 600 m on the Nûgssuaq Peninsula of West Greenland. The results reveal a record of two polarity transitions on the Disko Island and the existence of normal (N) and reverse (R) groups of lava flows on the Nûgssuaq Peninsula. A tentative correlation of the lava sequences from the two areas, on the basis of polarity of magnetization, has been suggested. Subject to the assumption of relatively uniform extrusion rates in northern Disko, a correlation of the comparatively thick sequence of reverse (R) polarity flows with the relatively long reverse epoch between the anomaly no. 25 (ca. 63 m.y.) and anomaly no. 24 (ca. 60.5 m.y.B.P.), has been attempted and an age estimate of the lava flows has been obtained.

The Early Tertiary paleomagnetic pole for Greenland, computed from stable remanent magnetic directions of the Disko lavas is located at 67.5 °N and 165 °W. This pole position and the one for contemporaneous lava flows on the Baffin Island of Canada have been used in testing the models proposed by various workers for a paleogeographic reconstruction of Greenland and Canada, involving a closure of the Baffin Bay – Labrador Sea. The results of this paleomagnetic test suggest the existence of an ocean basin in the area, prior to the eruption of Early Tertiary lava sequences on Baffin Island and West Greenland, and also that this ocean basin had a much wider extent in Pre-Tertiary times.

A model for the evolution of Baffin Bay – Labrador Sea has been suggested in the light of available geological/geophysical information about the region. It involves the opening of this sea in the Mesozoic and its partial closure during the Cenozoic as a consequence of the drift of Greenland in the northwesterly direction.

Des recherches paléomagnétiques ont été effectuées sur environ 250 échantillons de bloc orienté recueillis dans des coulées de laves du Tertiaire inférieur représentant environ 1200 m d'épaisseur sur l'Ile Disko et 600 m sur la Péninsule Nûgssuaq du Groenland occidental. Les résultats révèlent l'empreinte de deux transitions de polarité sur l'Ile Disko et l'existence de groupes normaux (N) et inverse (R) de coulées de laves sur la Péninsule Nûgssuaq. Un essai de corrélation des suites de laves des deux région, s'appuyant sur la polarité de la magnétisation, a été suggéré. Assujeti à l'hypothèse d'un taux d'extrusion relativement uniforme dans Disko septentrional, une corrélation de suites comparativement épaisses de coulées à polarité inverse avec l'époque inverse relativement longue entre l'anomalie numéro 25 (Ca. 63 m.a.) et l'anomalie numéro 24 (ca. 60.5 m.a. B.P.), a été tentée et un estimé de l'âge des coulées de laves a été obtenu.

Le pôle paléomagnétique du Tertiaire inférieur du Groenland, calculé à partir des directions

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magnétiques remanentes stables des laves de Disko est localisé aux 67 °N et 165 °W. Cette position du pôle et celle de coulées de laves contemporaines sur l'Ile Baffin du Canada ont été utilisées pour la vérification des modèles proposés par divers chercheurs pour une reconstruction paléogéographique du Groenland et du Canada, incluant une fermeture de la Baie de Baffin et de la Mer du Labrador. Les résultats de cette vérification paléomagnétique suggère l'existence d'un bassin océanique dans la région, avant l'éruption des suites de laves du Tertiaire inférieur sur l'Ile de Baffin et sur le Groenland Occidental, et aussi que ce bassin océanique était beaucoup plus étendue avant le Tertiaire.

Un modèle de l'évolution de la Baie de Baffin et de la Mer du Labrador a été suggéré à la lumière des données géologiques/géographiques disponibles sur la région. Cela implique l'ouverture de cette mer au Mésozoïque et sa fermeture partielle durant le Cénozoïque dû à la dérive du Groenland dans une direction nord-ouest. [Traduit par le journal]

Introduction

Considerable amount of paleomagnetic data on rocks of various ages has been obtained from North America and Europe and this has been used in attempting to outline the North Atlantic paleogeography. Since in all such reconstruction models Greenland is considered to have moved as a distinct crustal plate, reliable paleomagnetic data from Greenland are crucial. However, until recently, such data from Greenland have been rather sparse because of inaccessibility and difficulties in sampling.

Sequences of basaltic lava flows of Early Tertiary age are exposed on the western and eastern margins of Greenland and these formations together with the lava sequences on Baffin Island of Canada, Faeroes, and the British Isles have been considered as constituents of the Brito-Arctic volcanic province. Tarling (1967) reported the results of the first paleomagnetic study on the East Greenland Tertiary lavas and this was followed by a West Greenland result from lava flows on Ubekendt Ejland (Tarling and Otulana 1972). Kristjansson and Deutsch (1973) have recently reported another West Greenland result from lavas sampled in the southern part of the Disko Island.

We report here paleomagnetic results on lava samples collected from the northern part of the Disko Island and from the Nûgssuaq Peninsula (Fig. 1). Interpretation of these results together with those from Baffin Island of Canada (Deutsch *et al.* 1971) is discussed in relation to the evolution history of the Baffin Bay – Labrador Sea region.

Geology and Age

The West Greenland basalt province is made up of plateau basalts exposed on the Disko Island, the Nûgssuaq Peninsula, the Svartenhuk Peninsula and the Ubekendt Ejland (Fig. 1). The basalts extend from 70°N to 73°N and on the eastern side they disappear beneath the ice-cap (Munck and Noe-Nygaard 1957). Park *et al.* (1971) have recently suggested that they extend offshore for a considerable distance along the continental shelf in the Davis Strait and the total area covered by onshore and offshore basalts has been estimated to be about 55 000 km^2 (Clarke and Pedersen 1974).

The lava sequence on land has been classified

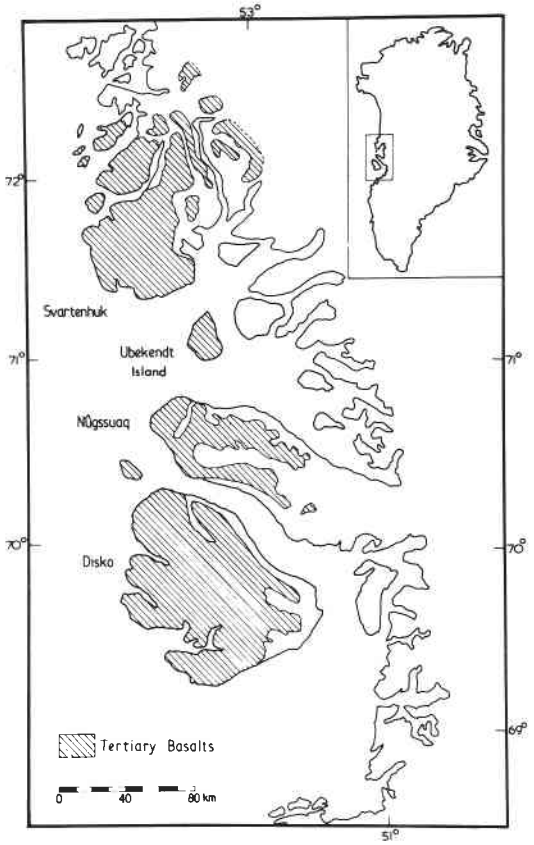


FIG. 1. The West Greenland basalt province.

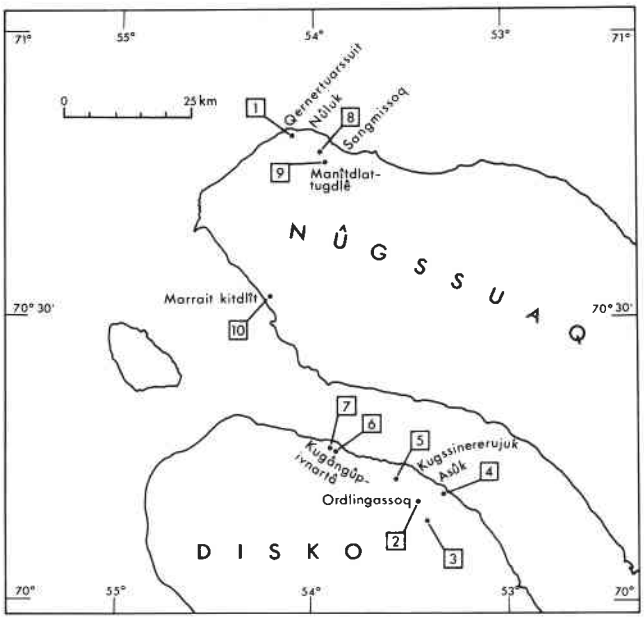


FIG. 2. Map showing the sampling localities on the Disko Island and Nûgssuaq Peninsula.

into three groups. The lower group of subaqueous pillow breccias is followed by a sequence of picritic basalts and olivine basalts; the top sequence, which is thickest, is made up of feldspar-phyric basalts (tholeiites). The thickness of the different groups of flows is variable.

The stratigraphy and structural setting of this region are described by Rosenkrantz and Pulvertaft (1969). The basalts are underlain by marine and limnic strata of Cretaceous–Early Tertiary age, which rest on the Precambrian basement. The youngest sediments are exposed in central Nûgssuaq and belong to the Agatdal Formation which is considered to be of Upper Danian age (Rosenkrantz 1970). The Upper Danian sandstone is overlain by pillow breccias and lava flows constituting the volcanic sequence. The main eruptive activity in the Disko–Nûgssuaq area is therefore considered to have commenced during the Upper Danian.

In the Nûgssuaq peninsula, block faulting accompanied by an antithetic fault system is very common and the lava flows have variable dips (Henderson 1969). Intercorrelation is difficult because of the complex nature of faulting. In contrast to this, most of the Disko Island, and in particular its northern part, is devoid of faulting; the lava flows are horizontal or have very gentle dips (Pedersen 1973).

Details of Sampling

In total 250 block samples representing 77 flows and 1 sill were collected from the Disko Island and Nûgssuaq Peninsula during the summer of 1971. The collection was made by K. A. Jørgensen and C. Mohr. On average three block samples were collected from each flow (site). In view of the erratic variations of the local magnetic declinations in Greenland and the strong magnetization of the basalts, the samples were oriented by using the sun-compass method as described by Creer and Sanver (1967).

Figure 2 shows the sampling localities. In northern Disko samples were collected from five profiles representing a cumulative thickness of about 1200 m. In Nûgssuaq three profiles were sampled on the northern coast and one profile on the southern coast near Marrait kitdlit. Available field evidence (Pedersen 1973) suggests that the lava flows in the northern Disko localities are horizontal or are dipping at very low angles ($\lesssim 3°$). However, this is not true for the Nûgssuaq localities and the amount of observed dip of the flows was recorded at each site.

During the summer of 1972 one site from the locality Kugangup ivnarta hill on the Disko Island was resampled by A. K. Pedersen for verification of results obtained on previously collected samples. Additional geological infor-

mation about the sampling areas can be found in Hald (1973) and Pedersen (1973).

Laboratory Measurements

Cylindrical specimens of diameter 2.5 cm and average height 2.4 cm were obtained from the block samples. Two specimens from each sample were used in measuring the natural remanent magnetization (NRM) on a PAR Spinner magnetometer. The stability of NRM directions was tested through alternating field demagnetization on an apparatus designed and built at the Paleomagnetic Laboratory in Copenhagen. This set-up consists of a specimen-holder with a capacity to hold eight specimens of standard size. The rotation of the specimen-holder takes place about two axes in the ratio of 13:19. The demagnetization coil has a capacity to produce peak-fields up to 3000 Oe at a current strength of 10 A. During demagnetization the field is reduced from its peak value to zero by using a rotating-drum potentiometer. A sliding contact moves at a uniform speed through the grooves on the drum surface, while remaining in continuous contact with the resistance strip embedded in the grooves. The field is gradually reduced to zero in 45 s.

To check the stability of NRM, one specimen from each site was subjected to alternating field (AF) demagnetization in successively increasing peak-field values of 60, 120, 180, 240, 300, 400, 500, 600, 700, and 800 Oe. Most of the specimens from Disko Island were found to possess stable NRM and their intensity of magnetization dropped to about 30–60% of the initial values on demagnetization in a peak field of 300 Oe, which was selected for batch cleaning of remaining specimens from all sites. The Nûgssuaq specimens also responded similarly to AF demagnetization and a peak-field of 300 Oe was adequate for obtaining the stable remanent direction which was rather far removed from the NRM directions.

The migration of remanent magnetization vector and decay of magnetization intensity in the case of two specimens (one normal and one reversely magnetized) from sites in localities 6 and 5 on demagnetization in peak-field strengths up to 800 Oe is illustrated in Figs. 3a and 3b.

Paleomagnetic Results

The stable directions obtained after AF demagnetization in peak-field of 300 Oe were used in computing the mean directions of remanent magnetization by using Fisher's (1953) method. The mean direction of each site was computed by giving unit weight to a specimen and the mean direction of a locality was obtained by giving unit weight to each of the site mean directions. The precision parameter k and the radius of the circle of confidence at 95% level were also computed in each case. Some sites were found to possess unstable magnetization. In the calculation of locality mean directions, we have omitted sites having a radius of circle of confidence greater than 15°. Seven out of the total 80 sites had to be omitted on this ground, two each from localities no. 2 and 5 (Disko Island) and one each from locality 6 (Disko) and localities 8 and 9 (Nûgssuaq).

Results from the Disko Island

The mean directions, statistical parameters and other relevant information about Disko localities are presented in Table 1. The results from locality 7 are presented in two groups as one of the sites was found to be reversely magnetized (R) while the other two overlying sites showed normal magnetization (N). An R–N–R polarity sequence is discernible in Table 1 from the results of localities 5, 6 and 7. This is particularly striking in view of the fact that stable normal (N) polarities have been so rarely identified in the Brito-Arctic Lower Tertiary belt and none had been reported previously from Greenland.

The paleomagnetic pole position calculated from the mean direction for northern Disko lavas is presented in Table 2. The pole position obtained by Kristjansson and Deutsch (1973) for 37 sites from southern Disko are also presented in this table along with the overall mean of Early Tertiary pole position by combining the results from a total of 81 sites from northern and southern Disko. The results obtained for the Paleocene lavas from East Greenland (Tarling 1967), Ubekendt Ejland (Tarling and Otulana 1972) from West Greenland (see Fig. 1), and from basalt flows at Cape Dyer on Baffin Island of Canada dated about 58 m.y. (Deutsch *et al.* 1971) are also presented in Table 2. The various pole positions are plotted in Fig. 4.

Ketelaar (1963) has studied the remanent magnetization of some Trap Diabase dikes from southwest Greenland. The pole position obtained from his results is presented in Table 2.

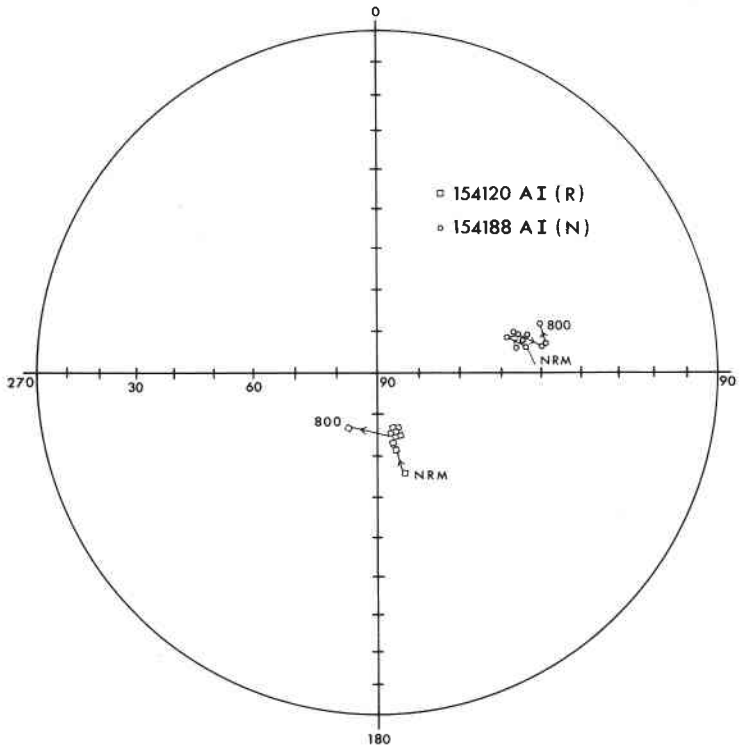


FIG. 3a. Change in the direction of NRM of two specimens, with magnetically stable reverse (R) polarity (locality 5) and stable normal (N) polarity (locality 6) on demagnetization in successive steps up to a peak-field of 800 Oe. Lambert equal area projection.

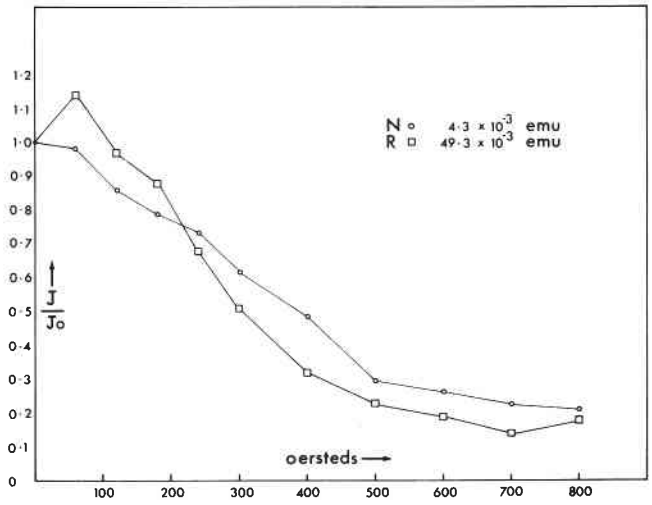


FIG. 3b. Variation of normalized magnetic intensity (J/J_0) with peak-field strength for the two specimens in Fig. 3a.

TABLE 1. Paleomagnetic directions from northern Disko

Locality	Coordinates	Altitude span	No. of sites	Stable remanent magnetic directions		Polarity	k	α-95
				D	I			
No. 3 1 km SE of Ordlingassoq	70°09′ N 53°25′ W	1105 m to 1450 m	15	147°	−74°	R	25.5	7.2°
No. 2 NE side of Ordlingassoq	70°11′ N 53°26′ W	840 m to 1190 m	14	154°	−73°	R	46.0	5.3°
No. 5 West wall of the valley of Manditlat Kugssinerssat	70°13′ N 53°33′ W	680 m to 365 m	15	123°	−73°30′	R	63.6	4.5°
No. 6 East side of the coast near Kugangup Ivnarta	70°16′ N 53°53′ W	325 m to 450 m	5	5°	+84°	N	6.6	24.3°
No. 7 West side of the coast near Kugangup Ivnarta	70°16′ N 53°53′ W	260 m to 300 m	2	348°	+51°	N	—	—
No. 7 same description as above	70°16′ N 53°53′ W	230 m	1	108°30′	−71°	R	—	—

D declination
I inclination
k – precision parameter
α-95 – Radius of circle of confidence at 95% level

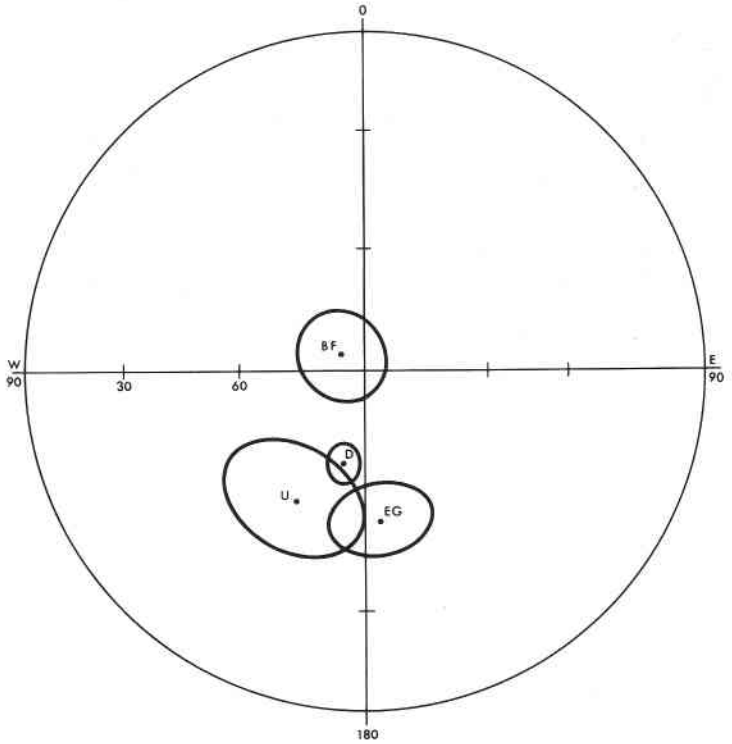


FIG. 4. Paleomagnetic pole positions for Early Tertiary lavas from Baffin Island (BF), Disko Island (D), Ubekendt Ejland (U) and East Greenland (EG). Lambert equal area projection with North pole at the pivot.

There is, however, considerable uncertainty regarding the age of these dikes.

Results from the Nûgssuaq Peninsula

The mean directions obtained for four localities in the Nûgssuaq Peninsula are presented in Table 3. The three localities (no. 1, 8 and 9) from the northeast part of the peninsula are found to be reversely magnetized and the single locality (no. 10) in picrite basalts exposed near Marrait kitdlit on the southwest coast was found to be normally magnetized. The normally magnetized basalts at locality no. 10 can be correlated with those at localities 6 and 7 on the Disko Island. The samples at localities 1, 8 and 9 collected from the tholeiitic group on Nûgssuaq (Hald 1973) have reverse polarity like the tholeiites from locality 3 (on North Disko) which represents the top section of the Disko sequence. It is, however, not possible at this stage to ascertain whether the two groups of lava flows represent the same polarity epoch.

The Nûgssuaq results show a large dispersion in the mean directions; however, this is reduced to a certain extent after correcting for local dips of the flows. The corrected mean directions are presented in Fig. 5.

As mentioned earlier, the tectonic history of the Nûgssuaq lava sequences is very complicated with several generations of faults and repeated antithetic faulting. As the amounts of absolute or relative rotations are not known with certainty, we have not calculated the mean paleomagnetic pole positions for these localities.

Significance of the Polarity Transitions in the Disko Lavas

We have mentioned previously (Table 1) the interesting finding of a normal (N) polarity zone in the predominantly reverse (R) polarity sequence in northern Disko. Of the two polarity transitions forming an R–N–R sequence, one is located at the base of the sequence at an altitude of 230 m and is a R to N transition. The reversely magnetized flow (at 230 m) is resting on the pillow-breccia sequence and therefore represents the beginning of the subaerial volcanism on the

TABLE 2. Paleomagnetic pole positions from Greenland and Baffin Island

Sl. No.	Formation and locality	No. of sites	Pole position Lat.	Pole position Long.	Radii of oval of confidence δp	Radii of oval of confidence δm	Reference
1.	Lava flows,* S. Disko, W. Greenland	37	62° N	169° W	8°	9°	Kristjansson and Deutsch 1973
2.	Lava flows,* N. Disko, W. Greenland	44	72° N	154°30′ W	5.5°	6°	This paper; mean of localities no. 5, 2 and 1 (Table 1)
3.	Disko Island,* W. Greenland	81	67°30′ N	165° W	4.9°	5.6°	This paper; mean of 1 and 2
4.	Lava flows,* Ubekendt Ejland, West Greenland	6	54° N	152° W	15°	18°	Tarling and Otulana 1972
5.	Lava flows,* Scoresbysund, E. Greenland	28	63° N	174° E	11°	14°	Tarling 1967
6.	Trap Diabase** dykes, Ivigut, SW Greenland	15	68° N	174° E	9.6°	12.5°	Ketelaar 1963
7.	Lava flows,*** Cape Dyer, Baffin Island, Canada	5	83° N	55° W	11.5°	12°	Deutsch *et al.* 1971

*Early Tertiary age
**age uncertain (Jurassic – Early Tertiary)
***considered to be of Early Tertiary age, although a single K–Ar result by a commercial laboratory (personal communication, E. R. Deutsch, 1974) has given a date of 70 ± 4 m.y.

TABLE 3. Paleomagnetic directions from Nûgssuaq

Locality	Coordinates	Altitude span	No. of sites	Mean stable remanent magnetic directions D	Mean stable remanent magnetic directions I	Polarity	k	α-95	Local structural dip	Corrected mean stable remanent magnetic direction D	Corrected mean stable remanent magnetic direction I
No. 1 valley, 2.5 km E of Kannisut	70°49′ N 54°05′ W	1 to 4 m	3	298°	−71°	R	23	17°	22° W	208°	−80°
No. 8 SSW of Sangmissoq	70°48′ N 53°57′ W	320 to 360 m 425 to 620 m	5	138°	−68°	R	39	10°	10° W	163°	−72°
No. 9 North of Manditlat Tugdle	70°47′ N 53°56′ W	400 to 550 m	2	191°	−78°	R	—	—	10° W	225°	−73°
No. 10 Near Marrait Kidtlit	70°31′ N 54°11′ W	100 to 125 m	9	285°	+51°	N	95	4.8°	22° W	284°	+73°

Legend as in Table 1

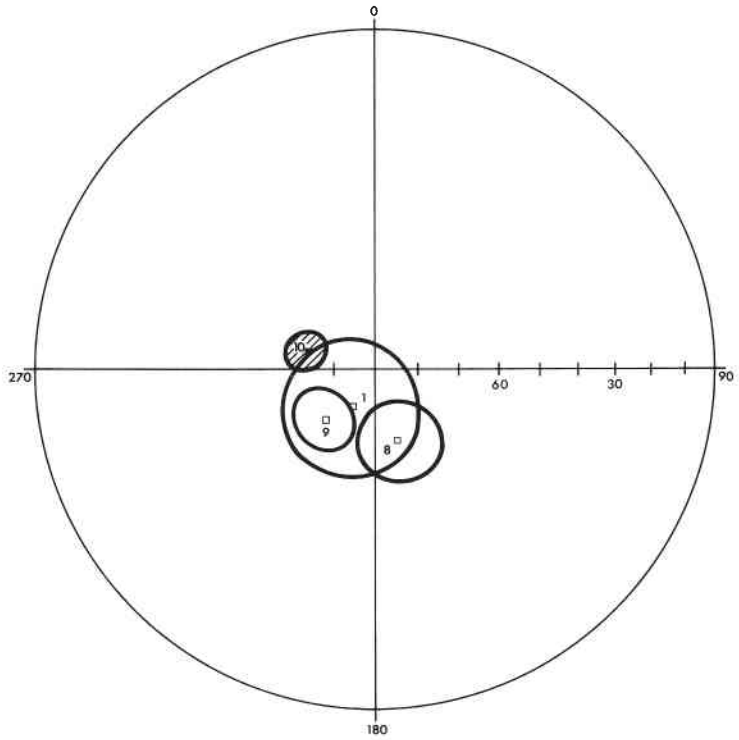


FIG. 5. Mean stable remanent magnetic directions for lava flows from four localities in Nûgssuaq peninsula. Locality 10 (indicated with hatching) plotted on the lower hemisphere; localities 1, 8, and 9 on the upper hemisphere. Lambert equal area projection.

Disko Island. The polarity transition recorded here has therefore a potential utility as a stratigraphic marker.

There is a conspicuous lack of radiometric age data for the Disko lavas. Determination of the absolute age of the Disko lavas by the K–Ar method has been attempted through whole-rock analysis of several samples, but the results so far have not been satisfactory (O. Larsen, personal communication 1974). Since this was written, Prof. E. R. Deutsch of Memorial University, Newfoundland has kindly communicated to us a single K–Ar result (obtained by a commercial laboratory) for one flow in southern Disko, that gives a date of 70 $\pm$ 4 m.y. The 70 m.y. figure is comparatively larger than most of the dates reported from other parts of the Brito-Arctic volcanic province. Thus the best control available at present for the dating of Disko lavas is that provided by the marine fossil evidence (mentioned previously) which places the beginning of volcanic activity in the Upper Danian (Lower Paleocene), *i.e.* 65–60 m.y. ago.

It might be very tempting to correlate the polarity reversals recorded in the Disko lavas with those constituting the geomagnetic time scale proposed by Heirtzler *et al.* (1968) from the analysis of marine magnetic anomalies. However, it tends to be very speculative. The Upper Danian epoch has not been assigned an age on the Phanerozoic time scale (Anonymous 1964) or on the geomagnetic polarity scale of Heirtzler *et al.* which is tied to the former. This makes it difficult to correlate the Disko polarity sequence (R–N–R) with a specific anomaly number on the Heirtzler *et al.* scale. Notwithstanding this difficulty, one could tentatively match the normal (N) polarity zone in the Disko sequence with the marine magnetic anomaly no. 25, on the argument that the comparatively long reverse epoch (during the period 60–65 m.y.B.P. on the Heirtzler *et al.* scale) between the anomaly no. 25 (63.28–62.75 m.y.) and anomaly no. 24 (60.53–60.01 m.y.) correlates better with the relatively thick (over 1000 m) pile of reverse polarity flows. This correlation is subject to the (speculative)

assumption that relatively uniform extrusion rates prevailed in northern Disko and that the flow of reverse (R) polarity at the base of the sequence does not represent a polarity *event* of short duration. If this assumption were valid, the 200 m thick sequence of normal (N) polarity flows would then correspond to the anomaly no. 25 (63.28–62.75 m.y.). This figure gives us an estimate of the (minimum) average eruption rate for the picrite basalts forming the lower part of the Disko lava sequence as about 400 m (in thickness) per million years.

Paleomagnetic Comparisons between Greenland and Northern Canada

Paleomagnetic data obtained from rocks of contemporaneous age from North America and Europe have been used by several workers (*e.g.*, Runcorn 1956, 1965; Hospers and Van Andel 1968; Deutsch 1969; Larsen and LaFountain 1970; Philips and Forsyth 1972) for reconstructing the North Atlantic paleogeography. All these studies led to a general picture of the reconstruction of continents bordering the North Atlantic that supported the continental drift hypothesis put forth by Wegener (1929).

Paleomagnetic comparisons across the Baffin Bay are potentially useful, for if Greenland separated from Canada as part of the opening of the North Atlantic, the Baffin Island and Disko areas should have been contiguous. Kristjansson and Deutsch (1973) have realized the importance of such comparisons, but as the proposed separation between the Cape Dyer and Disko areas is small (600 km) they suggest an estimated minimum sampling of 150 lava flows on each side of the Baffin Bay for making a statistically adequate test. The statistical error on the mean paleomagnetic pole position for the Disko area has now reduced considerably as a result of additional data reported in this paper (Table 2). On the other hand, nothing has been added yet to the Baffin Island data (based on only five flows collected from Cape Dyer) and hence the comparison between the two coasts is not greatly improved. Nevertheless, if one could assume the Disko and Cape Dyer poles to be representative of the two coast areas, then it is possible to carry out a paleomagnetic test for the various models proposed for the separation of Greenland from Canada and the consequent evolution of the Baffin Bay – Labrador Sea.

The basic assumptions for applying such a test to the paleomagnetic data for Baffin Island and West Greenland are: (1) The stable remanent magnetic directions in these rocks represent the direction of the paleoambient geocentric axial dipole field. (2) No polar wandering has taken place since the eruption of these lavas and their present relative juxtaposition is a consequence of continental drift. (3) The two formations are of the same age.

Validity of the first assumption for the Tertiary period has already been established through paleomagnetic data (Irving 1964; McElhinny 1973*a*). McElhinny (1973*b*) has recently shown from a vector analysis of paleomagnetic data for the six major plates that no significant polar wandering occurred during the Tertiary period. As to the third assumption, both the lava sequences are considered to have been generated during Early Paleocene and the floral assemblage in the sediments underlying them have strong similarities (Clarke and Upton 1971).

The results of application of the paleomagnetic test suggested by Deutsch (1969) to various models proposed for evolution of the Baffin Bay – Labrador Sea region are discussed below.

Model 1

In the Bullard *et al.* (1965) reconstruction model, Greenland is rotated counterclockwise to fit with Europe and the two together are rotated clockwise to fit with North America which is maintained at its present position. The resultant movement of Greenland can be expressed as a clockwise rotation through 18° around a pole of rotation at 70.5° N: 94.5° W (Wells and Verhoogen 1967). In Fig. 6 we have shown the counterclockwise displacement of the mean paleomagnetic pole position (67.5° N: 165° W) from D_1 to D_2 as a consequence of clockwise rotation of Greenland according to this model. We have used the method suggested by Deutsch (1969) on a Lambert equal area projection for this purpose. The pole of rotation was transferred to the pivot of the projection and the Disko (D_1) and Baffin Island pole positions were correspondingly transferred. The colatitude (θ) for the Disko pole was read on the projection and the angular displacement p of this paleomagnetic pole on rotation through R (18° in this case) was calculated from the relation: $1 - \cos p = \sin^2 \theta\,(1 - \cos R)$. The displaced pole position (D_2) was then located on the projection at an angular distance θ from the pivot and p from the

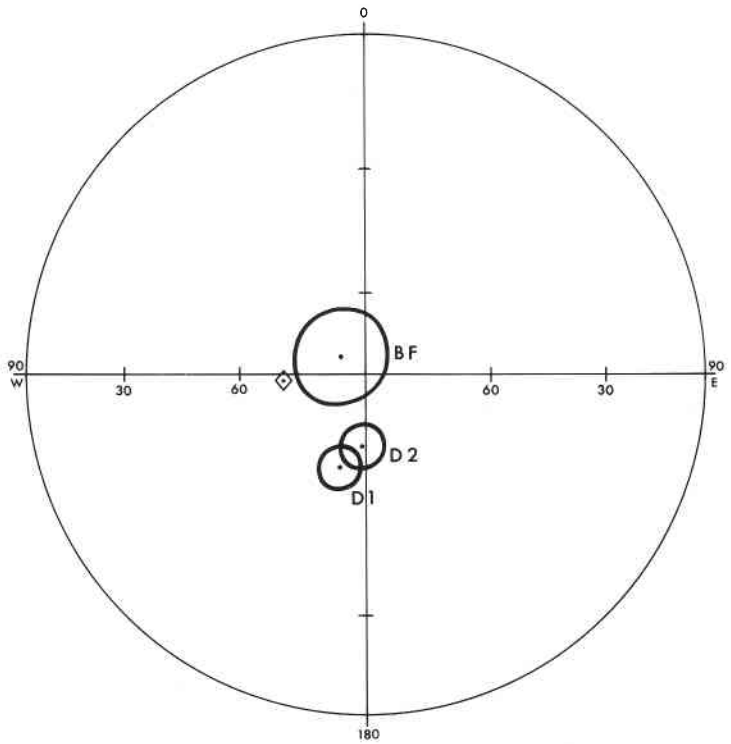


FIG. 6. Displacement of paleomagnetic pole for Early Tertiary lavas of Disko Island from original position D_1 to position D_2 on movement of Greenland according to model 1 in Table 4. The pole of rotation is indicated by a diamond.

original position (D_1). All the pole positions were then transferred back to their locations on a normal Lambert projection with the North pole at the pivot (Fig. 6).

It can be seen from Fig. 6 that the new displaced pole positions for the Disko Island (D_2) and Baffin Island (BF) remain widely separated despite the large oval of confidence associated with the latter. The possible explanations for this observation are:

(1) The two landmasses were not joined together at the time of eruption of the lava flows. This interpretation is contrary to the suggestion by Clarke and Upton (1971) that the Baffin Island and West Greenland basaltic provinces are fragments of a once continuous basalt plateau, now separated by continental drift and the intervening oceanic crust.

(2) The rotation poles and angles calculated for morphological fits are essentially mathematical entities, representing integral motion between two continents and they do not reveal the actual course of physical movements. In addition, it has been implicitly assumed, while using the Bullard *et al.* (1965) model for comparison of paleomagnetic data, that the present separation between the various continents around the Atlantic is the maximum separation since their break-up. As we shall see later, while this assumption may be valid for other landmasses, it may not be true in the case of present separation between Greenland and northeast Canada.

(3) We may also note that while the overall geometrical fit between Europe and North America obtained by Bullard *et al.* (1965) was fairly good, these authors point out that the western portion of Greenland did not fit well. Thus the present estimate of pole and angle of rotation for Greenland (Wells and Verhoogen 1967) based on the Bullard *et al.* fit between Europe–Greenland and North America may be associated with a certain degree of uncertainty.

Model 2

Le Pichon *et al.* (1971) have suggested a model for the opening of the Baffin Bay – Labrador Sea involving two main episodes. They estimated the pole of rotation and angle of rotation for the

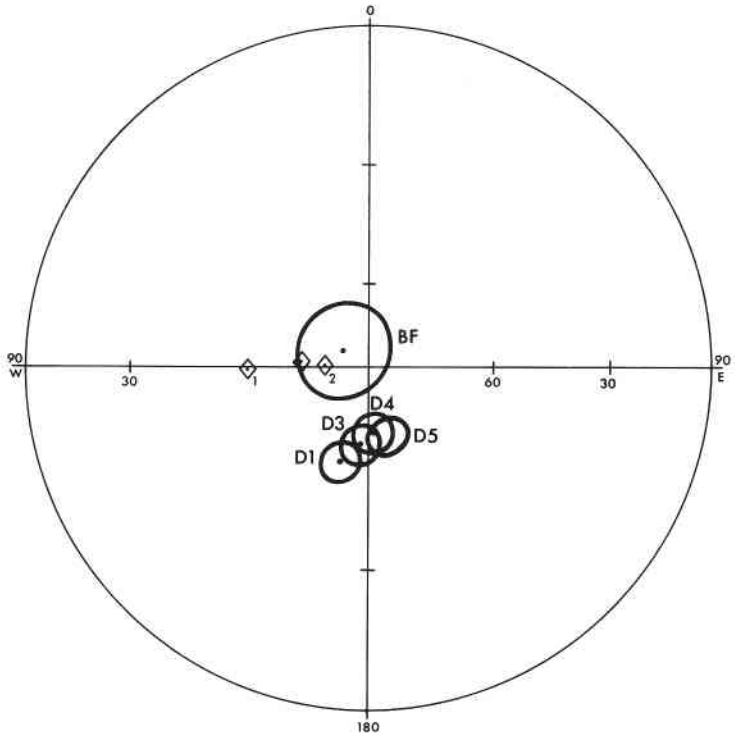


FIG. 7. Displacement of paleomagnetic pole for Early Tertiary lavas of Disko Island from original position D_1 to positions D_3, D_4, and D_5 on movement of Greenland according to models 2 and 3 in Table 4.

TABLE 4. Displacement of the Disko paleomagnetic pole according to various models discussed in the text

Model no.	Pole of rotation Lat.	Pole of rotation Long.	Angle of rotation R	Displaced position of Disko pole	Fig. No. and pole symbol
1	70.5° N	94.5° W	18°	73° N 178° W	6 : D_2
2 second phase	61.6° N	92° W	9.2°	73° N 176° W	7 : D_3
2 first phase	79.9° N	89.9° W	8.9°	74.5° N 176° E	7 : D_4
3	74° N	82° W	22°	73.5° N 171° E	7 : D_5
4	81° N	180° W	50°	78° N 150° W	8 : D_6

first phase from the trend of the Charlie (Gibbs) fracture zone located along 52° N in the Atlantic and deduced the values for the second (younger) phase of opening by subtracting the first set of values from those required for the Bullard *et al.* (1965) reconstruction model.

In Fig. 7 and Table 4 are shown the displacements of the Disko paleomagnetic pole from D_1 to D_3 and D_4 as a result of the two successive movements of Greenland according to this model. It is clear that the new Disko pole position (D_4) is still far removed from the Baffin Island pole position. This suggests again that the landmasses were further apart at the time of lava eruption (Early Tertiary) than they are at present. An alternative explanation is that the estimation of the pole and angle of rotation for the first phase of opening is associated with considerable uncertainty as pointed out by LePichon *et al.* (1971) and the estimation of the corresponding second phase values has the uncertainty associated with that for the Bullard *et al.* (1965) model discussed above.

Model 3

The analysis by Phillips and Forsyth (1971), within the frame of the Bullard *et al.* (1965) reconstruction model, of the paleomagnetic data

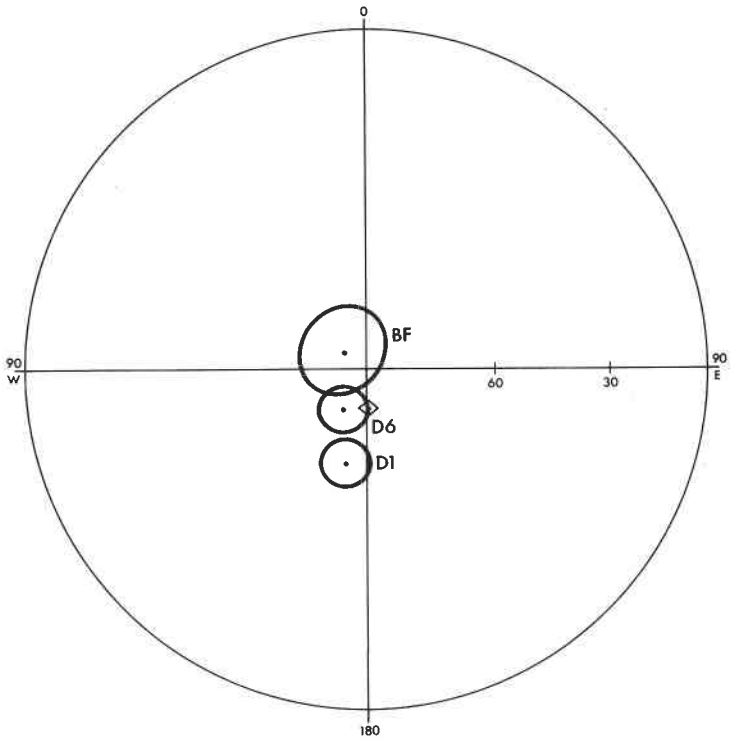


FIG. 8. Displacement of paleomagnetic pole for Early Tertiary lavas of Disko Island from original position D_1 to position D_6 on movement of Greenland according to model 4 in Table 4.

from continents surrounding the Atlantic has been already referred to by us. In addition, they also suggested an alternative empirical model according to which Greenland is rotated through 22° around a pole located at 74° N: 82° W. The resultant displacement of the Disko pole from D_1 to D_5 is also shown in Fig. 7. It is apparent that the amount of rotation suggested is not adequate to bring the Disko and Baffin Island poles close together.

Model 4

Clearly, none of the existing models for the evolution of the Baffin – Labrador Sea region can satisfactorily explain the large angular divergence between the Early Tertiary poles for Baffin Island and the displaced Disko pole positions. Theoretically, in a model satisfying paleomagnetic data, the two pole positions should coincide or at least their circles of confidence should overlap. Perusal of Figs. 6 and 7 shows that rotation of Greenland through a large angle around a pole of rotation located somewhere in the Arctic would be required for this purpose. We have found that with a pole of rotation located at 81° N: 180° W and a rotation angle of about 50°, the ovals of confidence of the Baffin Island and Disko Island poles would marginally overlap (Fig. 8). Starting with Greenland in its present position, such a large rotation would lead to the incongruous situation in which the Precambrian basements of Greenland and Canada would overlap. It is therefore clear that to accommodate such a large movement, Early Tertiary Greenland should be located further east.

Assuming that an ancient axis of sea-floor spreading was located midway in Baffin Bay, *i.e.*, approximately along 65° W, and that bilateral symmetrical spreading had occurred, we are required to move Greenland toward the east through about 25° from this ancient axis or through about 15° from its present position. Such a movement would bring the eastern shelf margin of Greenland approximately at the present location of Mohn's ridge, which is regarded as the site of separation between Norway and Greenland, 60–65 m.y. ago (Avery *et al.* 1968; Vogt *et al.* 1970). Evidence for northwestward movement of Greenland is also available

from the magnetic lineation pattern south of Iceland, where anomalies going back to no. 24 (60 m.y. B.P.) and parallel to the NE–SW trend of the Reykjanes ridge have been identified by Herron and Talwani (1972). In the Labrador Sea, south of Greenland, magnetic anomalies dating between 60 and 47 m.y. (anomalies 24 to 19) have been identified and these have a general trend in the NW–SE direction (Laughton 1971). These anomalies taken together with those of corresponding age parallel to the Reykjanes ridge, suggest that the resultant movement of Greenland was in the northward direction during Early Tertiary.

Looking at the paleomagnetic data from another perspective, we find that the mean paleomagnetic direction for Disko lavas (Az. 141°, Dip 71° upward) gives a paleolatitude of 56°, and this suggests that Greenland had moved northward across 15° of latitude since Early Tertiary. This represents only a part of the total translational movement which cannot be estimated through paleomagnetic data alone, since estimation of paleolongitudes is not possible. The mean value of declination (141°) however suggests that Greenland has rotated anticlockwise through about 40° since Early Tertiary.

We shall now examine the implications of the Model 4 after summarizing the available geological–geophysical information on the Baffin Bay – Labrador Sea region.

Geological–Geophysical Information on the Baffin Bay – Labrador Sea Region

At its northern extremity, the Baffin Bay – Labrador Sea is connected with the Arctic basin through the Nares Strait lineament (Fig. 9) which could have been formed in early Paleozoic times (Dawes 1973). Seismic studies in Baffin Bay suggest that its central portion is made up of oceanic crust of anomalous thickness, with a pile of sediments up to 7 km in thickness overlying the crust (Keen and Barret 1972). Similar thick sequences of sediments are found on the continental shelf margin on either side of Baffin Bay. Based on studies of marine sediments exposed on land and on the shelf of West Greenland, Henderson (1973) has suggested that the offshore sediments could be Jurassic or even older in age. There is no evidence in the Baffin Bay of a buried ridge or of a magnetic lineation pattern of the type expected to result from Cenozoic sea-floor spreading.

In the Davis Strait, the sediment thickness is about 1.5 km. The crust underneath has a thickness of over 20 km and is considered similar to the crust under Iceland. Magnetic lineation pattern is absent (Keen and Barret 1972; C. E. Keen *et al.* 1972) and there is no evidence for continuation across the Strait of Early Tertiary lava flows found off-shore on Baffin Island and West Greenland (M. J. Keen *et al.* 1972).

The geophysical studies carried out in the southern Labrador Sea (Le Pichon *et al.* 1971; Laughton 1971) suggest a sediment thickness of 2 to 3 km and magnetic lineations relating to anomalies no. 19 to 24 in the Heirtzler *et al.* (1968) scale. Several features of this old sea floor suggest that the larger deeper areas may be parts of an older Mid-Atlantic rift that extended northwest into a proto-Labrador basin (Vogt and Avery 1974).

In the northern Labrador sea, which we define here as the region between Greenland and Canada, south of Davis Strait and north of latitude 60° N, the sediment thickness is estimated to be about 1.5 km. Possible continuation in this region of a buried ridge identified in the southern Labrador Sea has been suggested (Johnson *et al.* 1973). Laughton (1971) has suggested the existence of magnetic lineations along the axis of this region and has taken the central anomaly to be no. 19 (47 m.y. B.P.). However, it appears that this interpretation results from the extrapolation of anomaly no. 19, tentatively identified by Vogt *et al.* (1969) in the southern Labrador Sea south of latitude 58° N. Hood and Bower (1973) have also suggested the existence of anomalies up to no. 13 (38 m.y. B.P.) in this region. Their identification of these anomalies is probably based on acceptance of continuation of anomaly no. 19 from the southern to the northern part of the Labrador Sea.

From the summary presented above, it appears that the Baffin Bay – Labrador Sea region has a complicated structure, probably comprising discrete tectonic units overlain by sediments of variable thickness. Unequivocal evidence for Cenozoic sea-floor spreading north of latitude 60° is not available at present.

Model for Evolution of the Baffin Bay – Labrador Sea

Based on our interpretation of the paleomagnetic data on Early Tertiary lavas from West Greenland and Baffin Island, and in the light of

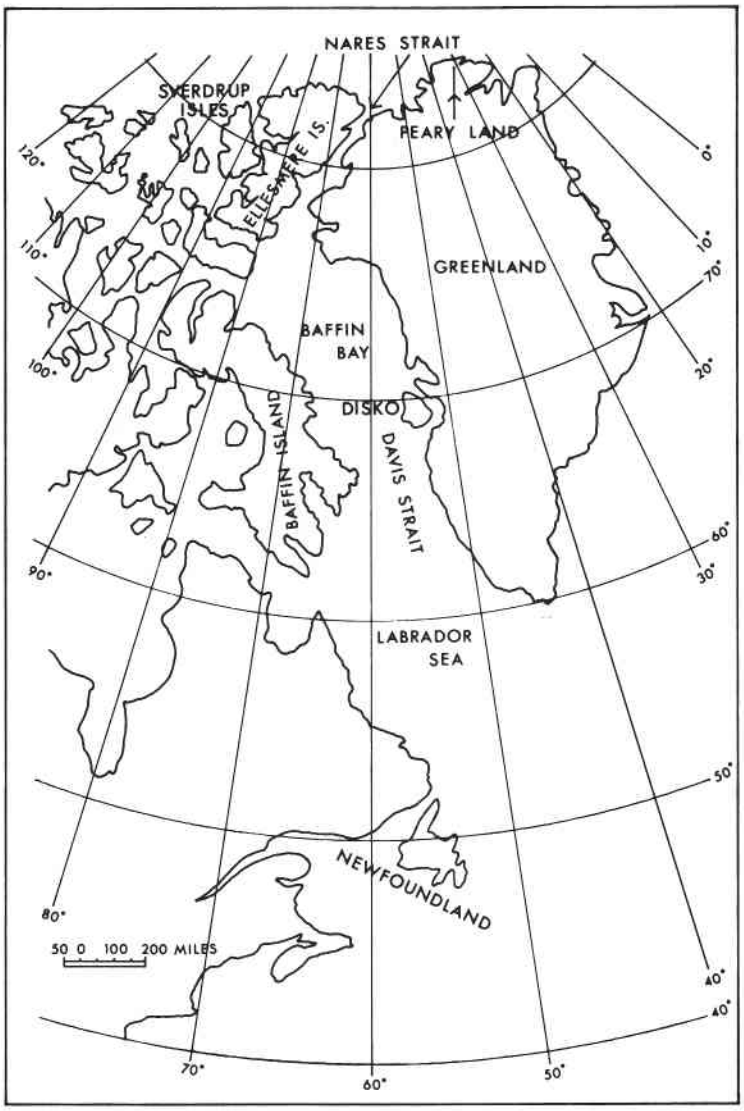


FIG. 9. Main morphological features of the Baffin Bay – Labrador Sea region.

the other available geological–geophysical evidence, we propose the following model for evolution of the Baffin Bay – Labrador Sea, north of latitude 60°:

(1) The Baffin Bay opened sometime during the Jurassic or even earlier, and sea-floor spreading continued until Late Cretaceous. Thick sequences of sediments were deposited in the sea basin and on the continental shelves of Canada and Greenland, which were separated by an angular distance of about 50° as a consequence of this spreading.

(2) Spreading ceased in the Baffin Bay – Labrador Sea, and the northeast branch of the Atlantic opened along the Mohn's Ridge and Reykjanes Ridge during the Late Cretaceous – Early Tertiary. Apparently at this time Greenland split from Europe, breaking off from Rockall Bank and Norway.

(3) Simultaneous spreading in the southern Labrador Sea and the Greenland–Norwegian Sea during the early Tertiary (corresponding to the period between anomalies no. 24 and 19) led to a resultant movement of Greenland in the northward direction.

(4) The Labrador and Norwegian Ridges became

extinct during the Middle Tertiary (about 30 to 40 m.y. ago according to Vogt and Avery 1974). Greenland then began to move northwestward as part of the American plate to occupy its present position.

Discussion of the Proposed Model

Our model for the evolution of Baffin Bay – Labrador Sea is based on the comparison of the Disko pole and the Baffin Island pole for contemporaneous lava flows. The implications of this comparison should be treated with due caution, for the latter result is based on only five flows sampled at one locality (Cape Dyer) and is statistically inadequate for comparison with the Disko result based on extensive sampling.

Despite this limitation the proposed model suggests a significantly large separation between Greenland and Canada as early as about 65 m.y. ago. Early Tertiary paleomagnetic comparisons between Greenland and northwestern Europe (results from the Faeroes and British Isles, Tarling 1970) also indicate displaced pole positions that do not agree with the simple Bullard *et al.* (1965) rotation model. E. R. Deutsch and L. Kristjansson (personal communication 1974) attribute this disagreement to the possibility that the main opening of the Norwegian Sea, beginning 60 m.y. B.P. (Vogt *et al.* 1970) postdates the time of major eruptions in West Greenland.

The proposed model is inconsistent with the suggestion (Clarke and Upton 1971), that the Baffin Island and West Greenland olivine basalt areas were once continuous and began rifting apart during the time of volcanism itself. However, on presently available geological and geophysical evidence (mentioned previously), together with the results of studies of magnetic anomaly lineations by LePichon *et al.* (1971), Pitman and Talwani (1972), and Vogt and Avery (1974), a mid-Mesozoic (or even older) rifting between Greenland and Canada is not at all improbable.

The working out of a more precise timetable of continental drift between Greenland and Canada poses several problems. Availability of reliable radiometric data for rocks on both sides of the Baffin Bay is crucial for a paleomagnetic comparison. Also, it is very important to obtain more paleomagnetic data from northeastern Canada, especially from Baffin Island. Additional paleomagnetic data from Greenland on Tertiary and Mesozoic rocks would be useful for resolving the various uncertainties in the history of evolution of the Baffin Bay – Labrador Sea. In particular, paleomagnetic data on Jurassic dikes from southwest Greenland, which are considered to have been formed during the rifting apart of Canada and Greenland (Watt 1969) would be of great potential value.

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